

MOD-2 MODEL QUESTIONS FOR PRACTICE

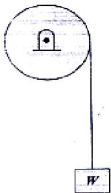
Q1. At a given instant, a shaft is rotating at 50 rpm about a fixed axis and 20 s later, it is rotating at 1050 rpm. Determine the average angular acceleration in rad/s^2 .

Q2. A flywheel of diameter 50 cm starts from rest with constant angular acceleration of 2 rad/s^2 . Determine the tangential and the normal components of acceleration of a point on its rim 3 s after the motion began.

Q3. A rotor decreases uniformly from rotating speed of 1800 rpm to rest in 320 s. Determine its angular deceleration and the number of radians rotated before coming to rest.

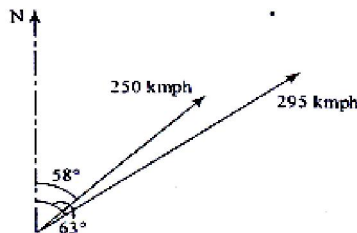
Q4. A bar pivoted at one end moving 5 rad/s clockwise is subjected to a constant angular deceleration. After a certain time interval, the bar has an angular displacement of 8 rad counterclockwise and has moved through a total angle of 20.5 rad . What is the angular velocity at the end of the time interval?

Q5. The drum shown in Fig. having diameter 800 mm is used to hoist the weight from a depth of 8 m. The drum accelerates uniformly from rest to 15 rpm in 1.5 s, and then moves at a constant speed of 15 rpm. What is the total elapsed time?



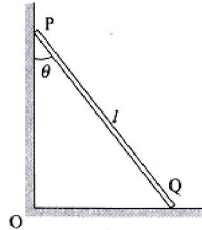
Q6. A flywheel of 550 mm diameter is brought uniformly from rest up to a speed of 350 rpm in 20 s. Find the velocity and the acceleration of a point on its rim 3 s after starting from rest.

Q7. After making correction due to magnetic deviation and variation, an aeroplane's true heading is computed as 63° with air speed 295 kmph. Some checkpoints indicate that the plane is making a true course of 58° at a ground speed 250 kmph [Fig. ---]. Determine the direction and the magnitude of the blowing wind.

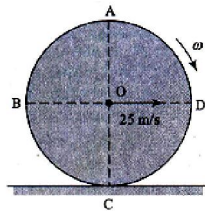


Q8. A car A travels along a straight road heading northwest at a speed of 20 kmph. Another car B moves along a straight road 60° south of west at a speed of 70 kmph. Determine the relative velocity of A with respect to B and that of B with respect to A.

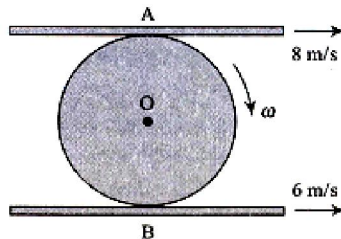
Q9. A bar of length l making an angle θ with the vertical, moves its foot point Q towards the right at constant speed V_Q as shown in Fig. Using instantaneous centre of rotation method, Determine the expression for angular velocity (θ') in terms of V_Q , l , and θ .



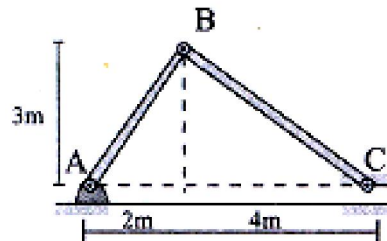
Q10. A roller of radius 2.5 m rolls without slipping along a horizontal plane as shown in Fig. The centre O has uniform velocity 25 m/s. Using instantaneous centre of rotation method, determine the velocity of the points A and B of the roller.



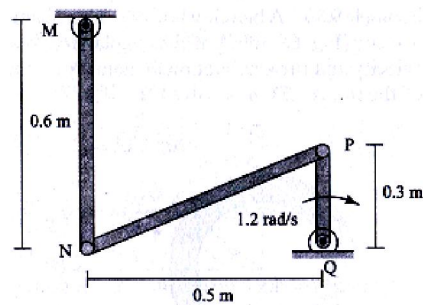
Q11. A roller moves within two conveyor belts, of which top belt moves at the speed of 8 m/s and bottom belt moves at 6 m/s as shown in Fig. Using instantaneous centre of rotation method, determine the linear and angular velocity of the roller if the diameter of the roller is 55cm.



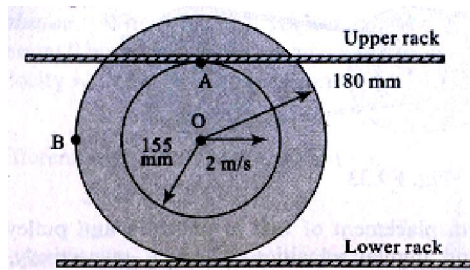
Q12. In the mechanism shown in Fig., B is a hinge joint and end C of member BC moves on a horizontal surface. Using instantaneous centre of rotation method, determine (i) the velocity of point C , and (ii) angular velocity of member BC . The member AB rotates clockwise about A at the rate of 3 rad/s.



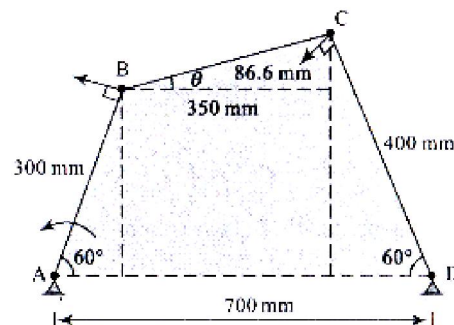
Q13. In a particular instant of moment, as shown in Fig., using instantaneous centre of rotation method, determine the velocity of N. Also determine the angular velocity of members PN and MN.



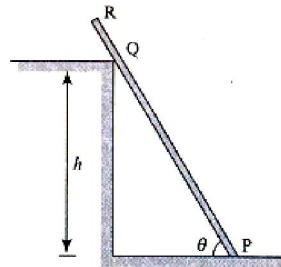
Q14. A double gear arrangement as shown in Fig. , rolls on a lower rack, which is stationary. The velocity of centre O is 2 m/s towards the right. using instantaneous centre of rotation method, determine the angular velocity of the gear. Also determine the velocity of the upper rack and point B.



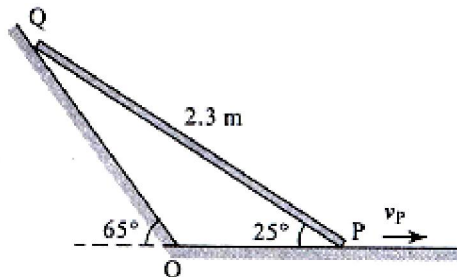
Q15. In the four bar linkage, shown in Fig, crank AB is rotating counterclockwise at 3 rad/s. Using instantaneous centre of rotation method, determine the linear velocities of points Band C and angular velocities of BC and DC.



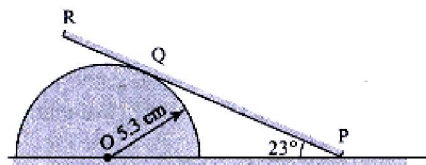
Q16. A slender prismatic bar PR supports at any point Q on the vertical wall as shown in Fig. If the point P has a velocity constant in value and directed horizontally, Using instantaneous centre of rotation method ,determine the expression for angular speed of the bar.



Q17. In the configuration of the solid bar PQ, shown in Fig., with $V_p = 2.4 \text{ m/s}$ constant, using instantaneous centre of rotation method , determine the angular speed of PQ and the velocity of end Q.



Q18. A slender bar PQ rests on a hemispherical disc of radius 5.3 cm, as shown in Fig. If end P moves horizontally at 3.2 m/s, using instantaneous centre of rotation method determine the angular speed of the bar.



Q19. A wheel is made to roll without slipping towards the right, by pulling a string wrapped around a coaxial spool as shown in Fig. Determine the velocity with which the string should be pulled(V_s), so that the centre of the wheel moves with a velocity of 13.5 m/s.

